LAMMPS CODE SPECIAL CASES EDITORIAL

Force Field Setup

1. Give charge to the atoms as per requirement.

```
None
set type 1 charge 0
set type 2 charge 1
```

2. Specify the mathematical formula for pairwise non-bonded atom interactions and for LJ interaction, provide the cut sigma value and the value of distance from which the long range interactions to be treated as electrostatic. Also mention the method with which the long range electrostatic interactions to be calculated and the accuracy needed.

```
None
pair_style lj/cut/coul/long 8 10
kspace_style pppm 1.0e-4
```

3. Then provide the LJ interaction parameters, ie epsilon, sigma, and LJ cutoff length along with mentioning the pairs. Also mention if the LJ curve is to be shifted to make the interaction after a certain distance equal to 0.
   Syntax: pair_coeff <type_1> <type_2> <Epsilon> <Sigma> <Interatomic Dist> <lj cutoff value>
   Lj cutoff value = lj/cut * interatomic distance.

```
None
pair_coeff 1 1 1.0 5.0 8.0 # Nanoparticle O
pair_coeff 1 2 1.0 3.0 3.0 # O-P
pair_coeff 2 2 1.0 1.0 1.0 # Polymer P
pair_modify shift yes
```

4. Then provide the weights for the LJ interaction and Coulombic interaction. This is basically the weights of how the atom pairs 1-2, 1-3, 1-4 interact with each other. 1-2 -> Atoms directly bonded to each other, 1-3 -> Atoms separated by two bonds (forming an angle), 1-4 -> Atoms separated by three bonds (forming a dihedral).

```
None
special_bonds lj/coul 0.0 0.0 0.0
```

5. Then mention the type of bonds and angles for example harmonic, which means that energy of bonds and angles increase quadratically like a standard ideal spring (both linear and torsional).

```
None
bond_style harmonic
angle_style harmonic
```

6. Then provide the parameters for the bond type and angle type, for example for bonds of harmonic type, we need to provide the information of the atoms involved in the bond, like O-O or O-P etc, the equilibrium length value and similar parameters, ie the torsional constant and equilibrium angle for bond angles.

```
None
bond_coeff 1 100 5.0     # O-O
bond_coeff 2 100 3.0     # O-P
angle_coeff * 5 180 # all angle types
```

NVT Simulation

The NVT stands for Number of particles, Volume, Temperature (target temperature). These three parameters are kept constant in this process. The particles are given random velocities. At regular time points, the energy of the system is calculated using the velocity of each particle and thus the kinetic energy. Thus, we have the kinetic energy calculated at a given time.

Also, the formula for the kinetic energy of a system at a particular temperature is given by the formula below

$$\langle E_{total} \rangle = \sum_{i=1}^{N} \frac{f_i}{2} k_B T$$

, where fi is the degree of freedom of the i-th particle. So, we got a particular temperature value, T. We had taken a target temperature, and we got a temperature corresponding to the current kinetic energy of the system. If both the temperatures match or have a difference less than the tolerance, the system settles, else the velocities get adjusted. Here, the important distinction is that the velocity profile is chosen as Gaussian or any other standard profile in the starting, but the profile is not maintained throughout the simulation.

For running NVT simulation, a **thermostat** is used. Thermostat is a mathematical algorithm that continuously monitors the temperature of the system and adjusts the atomic velocities to maintain the target temperature.

1. First, we need to define unit style, boundary type, and atom_style. Unit styles are nothing but the unit systems like SI, CGS, etc. Unit systems like SI, CGS are too big in scale for molecular simulation. Therefore, real unit system is used for MD simulations. Below are the scales of the real unit system.

Quantity	Unit
Distance	Ångströms (Å)
Time	Femtoseconds (fs)
Energy	kcal/mol
Temperature	Kelvin (K)
Pressure	Atmospheres (atm)
Mass	g/mol
Force	kcal/mol·Å

Then, we have to mention the periodicity of the structure in every direction, ie x, y, z. Below are all the options for the boundary.

Style	Full Name	What it Does
`p`	**Periodic**	Atom leaving one side re-enters from opposite side
`f`	**Fixed**	Atom hitting the boundary is deleted or stopped — hard wall
`s`	**Shrink-wrapped**	Box boundary automatically adjusts to fit atoms
`m`	**Shrink-wrapped with minimum**	Like `s` but never shrinks below initial size

In this case, period boundaries are used in all the directions, hence “boundary p p p”.

Then, the atom_style full means LAMMPS stores all the parameters associated with the particles of a system, including mass, charge, position, bonds, etc.

Then, we import all include the config file obtained from vmd and the forcefield to be used in the system.

```
None
units  real
boundary p p p
atom_style full
read_data initialsys.config
log log-min.txt
```

```
include forcefield.lmp
```

2. Specify the particles that are grouped, i.e., they remain fixed at their relative positions. Even if the centre moves, the grouped particles move by the same amounts.

```
None
group g1 id 1 2 32 62 92
```

3. The next step is to mention the dynamics of the particles as the NVT runs with time. If there are N particles, there are N*(N-1) / 2 pairs of interactions. Computation of these many interactions is not feasible. Therefore, we neglect particles out of the cut-off distance. For that, LAMMPS maintains a neighbour list (a pre-computed list of nearby atoms for each atom). This list also changes as atoms move. So, in every particular number of steps, the neighbour list is rebuilt. We also check if rebuilding the neighbour list for this step is even required.

The Below command says, “Rebuild the neighbour list in every single time-step, without delay from the starting time and in every step, check if rebuilding is needed or not. In one page, store 100000 atom-neighbour pairs, and every atom can have at max 5000 neighbours.”

```
None
neigh_modify every 1 delay 0 check yes page 100000 one 5000
```

4. Then we give velocity to each atom and the syntax for that is as follows:
command [group] [style] [temp] [random_seed] [rotation] [distribution].
Here the random seed is given 4928459. It can be any number other than 0.

```
None
velocity all create 1.0 4928459 rot yes dist gaussian
```

5. After the interactions of neighbours and velocities are mentioned, now we need to define the operation to carry out in different time steps. For that, **fix** command is used. The “fix” command applies the conditions given to it in the given order. The syntax is as follows:
fix [ID] [group] [operation] [variable] [start_val] [end_val] [damping_factor]
Here, the variable is temperature, start and end value both are 1K, and damping factor is 100 for all the atoms and for atoms of group g1 (here the atoms which are just bonded with the Silica nanoparticle along with the nanoparticle itself), it is 1.
Damping Factor is essentially the measure of how fast the thermostat reacts to the change in temperature. There are many types of thermostat. For running NVT, generally Nose-Hoover thermostat is used and that is mentioned using “fix nvt”. So, according to the Nose-Hoover’s relations,

$$m\ddot{\mathbf{r}} = \mathbf{F} - \zeta m\dot{\mathbf{r}}$$

$$\dot{\zeta} = \frac{1}{\tau^2}\left(\frac{T_{current}}{T_{target}} - 1\right)$$

the damping factor is inversely proportional to the time constant, Tau.

```
Small τ  → thermostat reacts very quickly → aggressive correction
Large τ  → thermostat reacts very slowly  → gentle correction
```

Here, for the atoms of g1, Tau value is small and for all others, Tau value is large. So, as a general rule of thumb,

- Large Tau for Flexible molecules with many DOF,
  Studying dynamic properties
- Small Tau for rigid and small systems.

None
```
fix 1 all nvt temp 1.0 1.0 100.0
fix 2 g1 rigid/nvt molecule temp 1.0 1.0 1.0
```

6. The algorithmic and technical part ends with the previous command. Now we want to monitor the system variables while NVT is running. For that, there are two specific commands.
   thermo - When to print
   thermo_style - What to print
   thermo [interval]
   thermo_style [style] [variable values to print]

None
```
thermo_style custom step temp press vol density pe ke etotal spcpu
timestep  0.005
```

7. Save the values of the steps in a file. Here states are saved after every 100 steps.
   Syntax:
   dump [ID] [group] [style] [frequency] [filename]

None
```
dump 1 all atom 100 trajmin.lammpstrj
```

8. After the LAMMPS code fils is ready, run it with the number of steps.
   run [no of steps]

None
```
run 500000
write_data after.config
print " ******* "
```

```
print " ****** finished energy minimization *******"
print " ******* "
print "Done"
```